IJRAR.ORG E-ISSN: 2348-1269, P-ISSN: 2349-5138

INTERNATIONAL JOURNAL OF RESEARCH AND ANALYTICAL REVIEWS (IJRAR) | IJRAR.ORG

An International Open Access, Peer-reviewed, Refereed Journal

IDENTIFICATION OF COUNTERFEIT PRODUCTS USING BLOCKCHAIN TECHNOLOGY

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ABSTRACT

Blockchain technology is used to authenticate products, handling the issue of counterfeit items in the supply chain. Examples include counterfeit goods like malicious counterfeit drugs, which can cause huge amounts of monetary damage and threaten public safety. The problem occurs when third-party distributors duplicate barcodes or even create fake products, which could result in a collapse of trust for online transactions. Blockchain solves the problem by removing the requirement of third parties and it implements a tamper-proof verification system. Barcodes and other product features are transformed into digital signatures and are saved in the Blockchain to ensure their integrity. Since blockchain verifies every transaction in multiple servers, any unauthorized modification of data is detected instantly.

KEYWORDS: *Blockchain, Decentralized, Counterfeit, Ethereum*

1. INTRODUCTION

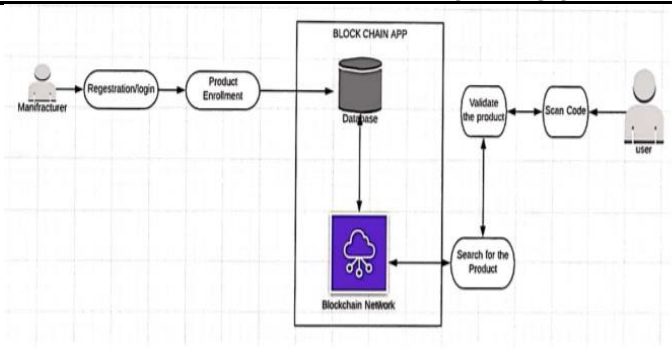
Fake products are a major issue in nearly every industry, but the supply chain, which has many distributors, is the most vulnerable. Fake products cost millions of dollars and even lives, especially in the medical industry. Legacy authentication systems easily depend on intermediaries and can be easily tampered with and abused by data. By storing product information and barcodes as digital signatures on a tamper-proof ledger, blockchain technology offers the secure, decentralized solution to this problem. No matter what you do, data will be detected immediately if altered. In this system, the Ethereum blockchain and smart contracts have been used to authenticate the products. Manufacturers help this process on their end by assigning an ID to products in the form of a QR Code, which the consumer scans to confirm authenticity. The process reduces risks of counterfeiting, safeguards, brand reputation, and increases customer trust.

2. EXISTING SYSTEM

Existing product authentication systems are built on centralized databases and third-party intermediaries. For example, barcodes and product labels verify authenticity. However, both are auxiliary and can still be hacked, tampered with, and subjected to cyberattacks, as data is stored on centralized servers. This increases the threat of fake goods making their way into the supply chain. One major limitation is reliance on intermediaries to validate transactions, which opens up the potential for fraud and misuse of information. Records are subject to the influence of malicious intermediaries who violate the security and authenticity of the records. This verification is often slow because it involves an intermediary, which delays authentication. In addition, it is very easy to duplicate barcodes and labels, which leaves us confused about which products are real and which are fake.

3. PROPOSED SYSTEM

We are utilizing blockchain technology that does not involve any third party, and verification will be performed by the software algorithm itself without any third-party involvement. To prevent counterfeit products, we are converting products details/barcodes into digital signatures and these digital signatures will be saved in Blockchain. As this blockchain supports tamper-proof data storage, nobody can hack or modify its data, and if by any chance its data gets modified then verification fail at next block storage. In Blockchain, all data will be saved by authenticating previous hash codes and if previous hash codes are not altered then data will be treated as original and then new transaction data will be added to Blockchain as new block. For every new storage of data, all blocks hash codes will be authenticated.



1. Manufacturer: The manufacturer logs into the application and registers the product using product details and its associated QR code on the blockchain to generate digital signatures.
2. User: The user can scan the product using the product QR code and verify the product, whether it is genuine or not.

SHA-256 Algorithm:

SHA-256 (Secure Hash Algorithm 256-bit) is a cryptographic hash algorithm that hashes input data and reduces it to a fixed 256-bit (32-byte) hash. SHA-256, created by the NSA, is one of the family members of the SHA-2 family of hash functions specifically designed to produce a one-way, irrecoverable, and deterministic hash, making it safe for digital signature and password hashing. SHA-256 works in a sequence of rounds of bitwise functions, modular addition, and compression functions to achieve collision and preimage resistance.

4. MODULES

1. Admin Application
2. User Application
3. Ethereum Smart Contract

MODULES DESCRIPTION:

1. Admin Application:
The Admin Application enables administrators to upload product information, create digital signatures from barcodes, and securely store them on the blockchain. It guarantees product authenticity by having tamper-proof records. Admins can view product details after adding them, and also, they can view the registered users.

2. User Application:
The User Application allows customers to check product authenticity by scanning QR codes/barcodes associated with blockchain records. Customers can access detailed product information and ensure they are purchasing genuine products. Discrepancies in digital signatures reveal counterfeiting, safeguarding consumers from counterfeit products.

3. Ethereum Smart Contract:
Ethereum Smart Contracts are computer programs that can execute automatically and securely handle transactions and product authentication on the Blockchain. They validate data integrity automatically and resist unauthorized changes. Through the removal of intermediaries, smart contracts increase security, transparency, and efficiency in product authenticity verification.

5. CONCLUSION

The use of blockchain technology for product authentication is a strong solution against counterfeit products within the supply chain. The data collected cannot be altered, thus making it a fit and proper technology for use in industries that require a high level of security. Through the encryption of product barcodes into digital signatures and the safe storage of them on the blockchain, the platform secures the genuineness and integrity of goods and ensures prevention from fraudulent operations.

6. ENHANCEMENT

Beyond supply chains, the future of blockchain technology application in identifying fake products is positive in all industries. In the pharmaceutical industry, it can help in the identification of the authenticity of drugs and avoid counterfeit drugs that may pose dangers to people's lives. In the luxury goods context, blockchain can assist in establishing product provenance to combat counterfeiting and brand and consumer protection.

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